

GT-B159 Lab Setup Instructions

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This document provides an overview of how to startup and use the UR10e robot arm in GT-B159.

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1 Things To Be Careful Of

These will make more sense after reading all of the instructions, but are important enough they are listed up front as well.

1. Always create a new conda environment for yourself or your group
2. You NEED our custom version of urx, you cannot use the one straight off the internet
3. Keep the e-stop in arms reach
4. Keep your common sense about you
5. Ensure the camera buffer is being emptied if you are using the camera

2 Hardware

2.1 Startup Procedure

1. Press the power button on the top right of the Control Panel.
2. Press the red circle in the bottom left.
3. Press ON. Wait a moment. Press START. You will hear the breaks release. Press Exit in the bottom left.
4. Ensure the mode in the top right is set to "Automatic" or "Local Control" instead of "Remote Control".
5. Press Load Program on the left.
6. Select the wait.urp program, and press Open. wait.urp should now appear as the program on the left. Hit the Play button. It should now say the program is Running.

7. Set the mode in the top right to be "Remote Control" instead of "Automatic".
8. The robot is now ready to receive commands from Nyx.

2.2 Shutdown Procedure

1. Select the green button in the bottom left. Hit off.
2. Press the power button on the top of the Control Panel, then select Power Off.

2.3 Reset after E-Stop

1. Release the E-stop by twisting it and pulling outwards.
2. You can select Not Now.
3. Press the yellow circle in the bottom left, and then hit START. You will hear the breaks release. Press Exit in the bottom left.
4. If you need to move the robot to a new safe starting configuration:
 - (a) Enter Automatic/Local Control mode in the top right.
 - (b) Select the Move panel in the top left.
 - (c) In the center of the screen, select either Home or Freedrive. Then press and hold the button that appears to either send the robot home, or to allow it to be moved by your motions.
 - (d) Once in a safe configuration, release the button and exit the Move menu.
 - (e) Enter Remote Control mode in the top right.

2.4 Ethernet Reconnection

To reconnect if UR attached to Vulcan or the NUC (and not Nyx), follow the below. This is applicable if the ethernet cord coming from the robot is not attached to the TP Link router seated by Nyx but is on the Franka's side of the room or attached to the NUC.

1. Take the gray ethernet cord and put it into the TP link router. If the ethernet cord was attached to the NUC, you can likely skip the following steps, as it should already be on the appropriate network.
2. On the Control Panel, enter Manual mode in the top right (if in Remote Control you have to go to Automatic first to have access to Manual) - you need a password, "123" will suffice.
3. Hit the three horizontal lines in the top right, then select Settings $\downarrow$ System $\downarrow$ Network.
4. Enter the Admin Password, which is written on the control above the IP address, written in green, all lowercase. Select Unlock.

5. Enter the following details, then hit apply:
 - IP Address: 192.168.0.2
 - Subnet Mask: 255.255.255.0
 - Default gateway: 192.168.0.1
 - Preferred DNS server: 0.0.0.0
 - Alternative DNS server: 0.0.0.0
6. Hit Exit in the bottom left, then return to Automatic or Remote Control mode via the top right.

2.5 Opening/Closing the Gripper via the Control Panel

1. Enter Manual Mode via the same steps as earlier.
2. Select Installation in top left.
3. Select URCaps > Gripper
4. Select E-open / E-close. If this does not appear, hit Scan first.
5. Select Run in the top left.

3 Software

- Use the whiteboard credentials (top left) to log in to the guest account.
- You can download files within a folder you create for your team in '~ /courses/' or '~ /research/' depending on why you are in the lab. If you are in a course, please use the subfolder with your course name and assignment (e.g. '~ /courses/me235/final_projects').
- Please create a new Conda environment in order to manage all dependencies without disrupting those of another group. For instance, for ME235 Lab 3, we used OpenCV version 4.13 and Numpy 2.2.6, if you would like similar specifications.
- DO NOT overwrite the installed URX library package in environment urx_test with the command 'pip install urx'. The version available via pip is NOT functional, and requires manual adjustments. Instead clone "Caltech-AMBER/python-urx.git" from github and install with "pip install ."
- To get the software that controls the gripper, you can use "pip install pyrobotique"

4 Camera Info

- Ensure the camera resolution is set to 1536x2304 when you create a camera object in OpenCV.
- Set the buffer length to one with `'self.cap.set(cv2.CAP_PROP_BUFFERSIZE, 1)'` if `self.cap` is the `VideoCapture` object.
- Run `'v4l2-ctl -list-devices'` once the camera is plugged in. There will likely be two devices listed with `'dev$NUM$'`. Use `$NUM$` as the camera index when creating the camera via `VideoCapture`.
- Ensure you are getting the latest camera picture by emptying the buffer.
 1. In your file that includes the camera's `.read()` function, import the `'threading'` library.
 2. Create a thread (normally as an object in your class structure) using `'self.thread = threading.Thread(target=self._update, daemon=True)'` and `'self.thread.start()'`.
 3. Use the thread's `_update()` function to constantly empty the buffer and keep the latest frame in a local copy called `self.frame`.
 4. Now whenever you need access to the latest frame, use `self.frame` instead of the camera's `.read()` function.

5 Remote Nyx Connection

- `ssh USERNAME@131.215.101.244`
- Use Port 22 if a port needs to be specified